بسم الله الرحمن الرحيم

Allergy

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Allergy

Definition of allergy

- **Allergy** is an altered reactivity to the tissue of certain individual, caused by exposure to an agent **(allergen)** that is innocuous to other persons.

- Or **altered reactivity** on second contact with allergen.

- **Allergen is an antigen** that can elicit allergic symptoms.

Route of administrations :-

A) Inhalation

B) Ingestion

C) Injection

D) Contact with skin

Classification:

Hypersensitivity reactions are best classified on the basis of the mechanism involved in their production.

a. Immediate allergic reactions include types I, II, III

b. Delayed allergic reaction include type IV

Prevention of drug allergy:

It is impossible to prevent drug allergy but rather to minimize the risks.

1) Careful history concerning:

- Previous drug reaction.
- Hay fever, asthma or any allergic reaction for the patient.
- Family history of allergic reaction.

2) Taking into consideration cross allergy. Patient may be allergic to different drugs that share the same basic structure e.g. penicillin and cephalosporin's.

3) Oral route is the safest and i.v. route is the most dangerous so avoid i.v. injection as far as possible.

4) Patients liable to develop allergic reactions should be observed for 30 minutes after parental administration.

(serious reaction occurs within 30 minutes).

5) Dentists should be equipped to deal with allergic reaction with **the suitable medication.**

(preload syringe of adrenaline and good medical information's)

6) The patient should be informed about the drug to which he was found to be allergic.

7) **When confirmed allergic;** avoid exposure to:-

- **Penicillin allergy** is prevented by utilization of erythromycin
- **Aspirin allergy** is prevented by utilization of acetaminophen
- **Methyl methacrylate monomer** allergy is prevented by not employing acrylic resins,

Drugs used to treat hypersensitivity:

(1) Adrenaline:

Action.:

The first line of treatment in emergency situation as anaphylactic shock, bronchial asthma and angioedema.

It produces:

1- Bronchodilator.

2- Vasoconstriction (mucosa , skin & viscera)

3- Decrease capillary permeability, **thus reduce edema**.

In addition, it **acts within few minutes** leading to increase in blood pressure and stimulation of heart muscle. Such actions are **life saving** in particular with anaphylactic shock.

Dose:

- 0.5 ml (1/1000 conc.) i.m. or s.c.
- **Re-administrated after 5 minutes** if there is no improvement with maximum dose of 1.5 ml.
- **Never to be given i.v.** to avoid **ventricular fibrillation.**

- **(2) Corticosteroids:**

Indication:

1. Anti-inflammatory
2. Immunosuppressant
3. Antilymphocytic cytotoxic in oncology
4. In the management of allergic diseases.

Mechanism of action:

• **Cortisone enters the cells** by passive transfusion [Lymphocytes, Monocytes, Macrophages & C.T cells].

• It combines with **cytoplasmic receptor**.

• The **complex will enter the nucleus** which will be stimulated to produce mRNA to produce **Lipocortin**.

• Action of Lipocortin

1) Anti-inflammatory action

- **Inhibits phospholipase enzyme** & so interfere with inflammatory mediator (prostaglandin, leukotrienes,) release i.e. posses anti-inflammatory action via stabilization of biologic membranes and capillaries.

2) Immunosuppressive function:-

A- **Suppression of cell mediated response**.
(inhibition of production of IL 2)

B - **Interfere with the function of macrophage**, since it inhibit

- Antigen presenting function.
- Secretory function (IL 1, IL 6 , PG & TNFa)
- Chemotaxis
- Phagocytic function : steroid impair the function of C3b &Fc receptors of macrophages.

- The immunosuppressive function is mediated via **lymphocytopenia& monocytopenia.**

3) In small dose steroid impair T cell function, while in high it impairs B cell function.

4) Redistribution of blood cells resulting in monocytopenia, lymphocytopenia and increase in Neutrophils.

Effect on blood cells	**Decrease Eosinophils** **Decrease basophiles** **Decrease monocytes – monocytopenia** **decreased T. lymphocytes** **& B. lymphocytes** **Natural killer cells resistant to GC**	**Neutrophils increased** **Neutrophilia**
Mechanism	**• Increase Monocytes & T. lymphocyte (CD4) migration to bone marrow, spleen & to lesser extent to LN.**	***Increase in intravascular half life of Neutrophils** **•Increase in mobilization of Neutrophils from bone marrow.** **•Decrease in local margination , migration and accumulation at site of inflammation.**
Action	**Prevent cells from encountering the antigen.**	**Decrease sign of inflammation**

(3) Antihistamines:

Antihistamines are of no value in emergency treatment since:

1) They are slowly acting drugs.
2) They antagonize one mediator only **(histamine).**
3) They restore the vasomotor tone with out producing potential vasoconstriction.

They include:

a- Promethazine Hcl

- **25** - 50mg oral or i.m. every 24 hours
- It is long acting and sedative (8-12 hours), given at night.

b- Mepyramine (50 mg tablet)

- 100 - 200 mg daily in divided dose.

c- Pyribenzamine (50 mg tablets)

- I - 2 tablets 3-4 times daily

Type of reaction	Mechanism
Type I:Anaphylactic (immediate)	Free antigens bind to IgE on mast cells and basophils, causing release histamine,
Type II:Cytotoxic (antimembrane)	IgG or IgM activate Complement bind to antigen on cell membranes → cell damage or phagocytosis
Type III:Immune complex disease	Persistance of Ag/Ab complexes may activate the complement → inflammation and tissue damage, vasculitis
Type IV: Cell mediated (delayed hypersensitivity reaction))	Antigen activates sensitized T cells to become cytotoxic and release of lymphokines that stimulate other leukocytes

Type of reaction	Mediators	Histopathology
Type I:Anaphylactic (immediate)	• IgE • Mast cell • Basophilisls	• Mast cell degranulation • Eosinophils • Edema • Vasodilatation
Type II:Cytotoxic (antimembrane)	• IgG • IgM • Complement	• Damage to target cells
Type III:Immune complex disease	• IgG • IgM • IgA	• Acute inflammatory reaction. • Neutrophils
Type IV: Cell mediated (delayed hypersensitivity reaction))	• TH1 • T cytotoxic • Activated macrophages • Lymphokines interleukinesuh	• Perivascular inflammation • Mononuclear cells • Fibrin • Caseation and necrosis in TB

Type of reaction	Examples
Type I:Anaphylactic (immediate)	1 - Generalized anaphylaxis Analphylactic shock 2 - Localized anaphylaxis a. Angio-edema b. Bronchial asthma c. Hay fever d. urticaria
Type II:Cytotoxic (antimembrane)	a. Autoimmune hemolytic anemia b. Thrombocytopenia. c. Erythroplastelies fetalies
Type III:Immune complex disease	a. Serum sickness b. Arthus reaction c. L.E
Type IV: Cell mediated (delayed hypersensitivity reaction))	a. Contact dermatitis b. Graft rejection c. Pulmonary T.B

Type I allergic reaction

- **Basis of reaction:**
- First contact with allergen , results in the formation of IgE by plasma cells via interaction between macrophages, TH2 and B cells. IgE circulates in the blood stream and binds ;

- More commonly to **mast cells**
- Less commonly to eosinophils, macrophages

& **sensitize them.**

- **Next contact with allergen** results in binding and cross linking of the antigen to the IgE on the mast cells, eosinophils and macrophages and many mediators will be released.

Mast cell and basophile granules are groped as:-

A – Performed granules

B - Synthesized substance from cell wall

A – Performed granules

(1) Histamine:

Action: Present as preformed mediator in sensitized mast cells and basophils.

A) Increase capillary permeability

B) Smooth muscle Contraction (bronchi)

C) Stimulate gastric acid, bronchial & nasal secretion

(2) Heparin

- **Anticoagulant**
- **Activate C1**

(3) Protease enzyme mainly tryptase

- Increases vascular permeability
- Activate C3

(4) Platelet activating factor

- Activation & **aggregation of platelets**
- **Vasodilatation of B.V**. and increase vascular permeability
- **Activation & degranulation** of netrophils and eosinophils
- **Activate complement**
- Production of **prostaglandin & leukotrienes**

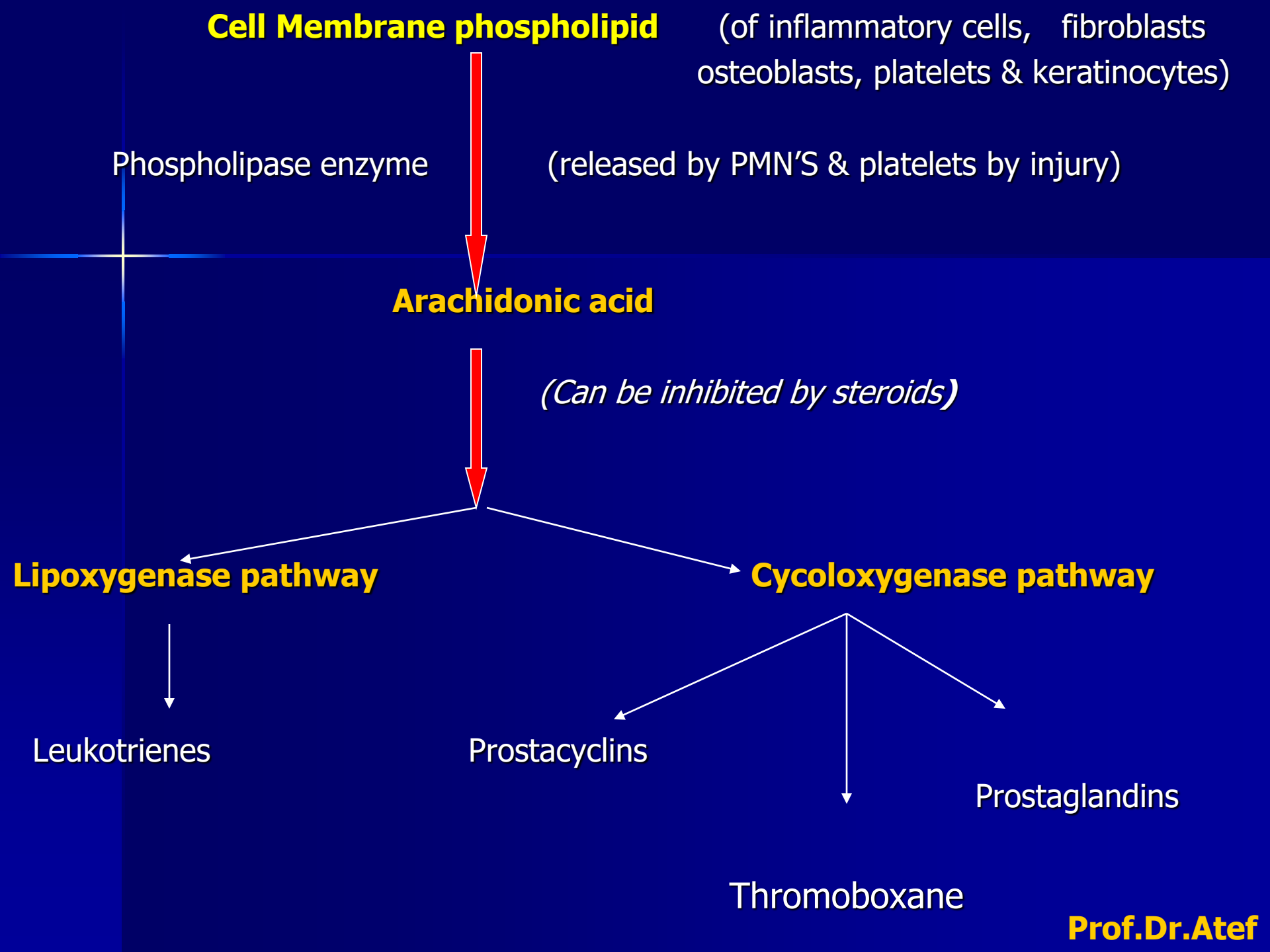
(5) Eosinophils Chemotactic factor

(6) Neutrophils Chemotactic factor

B - Synthesized substance from cell wall

- Generated in sensitized **cell from membrane** lipid following binding with specific allergen.

- **Arachidonic acid is metabolized** to produce
  - Prostaglandin,
  - Leukotrienes
  - Platelets activating factor.


Cell Membrane phospholipid
(of inflammatory cells, fibroblasts osteoblasts, platelets & keratinocytes)
Phospholipase enzyme
(released by PMN'S & platelets by injury)
Arachidonic acid
(Can be inhibited by steroids)
Lipoxygenase pathway
Cycoloxygenase pathway
Leukotrienes
Prostacyclins
Prostaglandins
Thromoboxane
Prof.Dr.Atef

Mediator	Main Source	Action
Prostaglandin	Neutrophils Eosinophils Monocytes platelets	Vasodilation increase vascular permeability pain bronchial mucosal edema
Leukotrienes	Neutrophils mast cells basophiles	vasodilatation increase vascular permeability contraction of smooth muscle induce adherence & chemotaxis Mucous hyper secretion Bronchial mucosal edema
Platelets activating factor PAF	Mast cells Platelets	Aggregation of platelets chemotactic to white blood cells. Increase vascular permeability Vasodilatation & broncho-constriction

- This type of **hypersensitivity reaction is immediate** where clinical manifestations appear within half an hour, the more rapid the clinical feature the more severe the reaction.

Type I reaction can be transferred by **injection of serum containing IgE into the skin** (passive cutaneous anaphylaxis).

The **antibody will bind to the mast cells** and remain fixed up to 4 -5 days in skin.

Subsequent injection of the specific antigen will produce wheel and flare reaction **(Prausintz - kustner test).**

Examples of Type I allergic reaction

(a) Generalized anaphylaxis

e.g. Anaphylactic shock

(b) Localized anapyhylaxis:

- Bronchial asthma
- Angio edema
- Hay fever
- Urticaria

Anaphylactic shock

Definition:

- It is the most dramatic and **dangerous reaction** that may follow administration of foreign protein or drug in sensitized individuals.

- **The reaction is acute** with simultaneous involvement of CVS, RS, GIT & skin.

- The reaction usually starts within 30 minutes (may be 30 second) after injection of a drug.

- **The quicker the onset , the more severe the reaction.**

Mechanism:

type I reaction resulting in:-

-Generalized vasodilatation of arterioles particularly the mesenteric vessels

-Increased capillary permeability.

Clinical features:

-The reaction usually starts **within 30 minutes**.

- The **quicker the onset**, the **more severe the reaction.**

- **Starts as:**

1) **Paraesthesia and edema** of the face.
2) **Coldness** of hands and feet.
3) **Wheeze** due to bronchospasm.

- **In severe cases:**

1) **Rapid severe fall** of blood pressure.
2) Pallor and sweating.
3) Fast thready pulse.
4) **Loss of consciousness**

- **Later**

1) The patient is **pulsless.**
2) **Cyanosis** from inadequate oxygenation.
3) **Death** may occur within five minutes.

- **Management:**

1 .Lay the patient flat with legs raised.

(This position improves venous return from the body peripheral regions to the heart and increases blood flow to the brain.)

2.Adrenaline. IM epinephrine is administrated: 0.5mg of 1:1000 of epinephrine solution. Re-administrated after 5 minutes if there is no improvement with maximum dose of 1.5 ml in 15 minutes.? **cardiopulmonary resuscitation.**

3 .Oxygen & monitoring of vital signs until an improvement in patient status is noted (blood pressure and pulse rate at carotid artery at least very 5 minutes).

N.B. These 3 steps **(position, adrenaline and oxygen)** are the only steps during the life-threatening phase of anaphylactic shock.

4. Hydrocortisone sodium succinate 200 mg I.V.

***5*.Antihistaminic.** 50 mg of diphenhydramine or 10 mg of chlorpheniramine IM., which will be followed by administration of 50 mg of oral diphenhydramine or 10 mg of oral chlorpheniramine every 6 hours for 24 hours.

6. Call ambulance.

- **N.B.** -The **4th** and **5th** steps are cortisone and histamine blockers (I.M. or if possible I.V).

-They are not administered during the acute phase of the reaction.

- **Hydrocortisone and antihistamine are not necessary administered during the acute phase of the reaction**, their main function

A - **prevent relapse of symptoms** (due to the release of the synthesized substance which will be formed later from the cell wall of mast cell and basophils)

B - **to obviate the need for continuous administration of adrenaline.**

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Angioedema

Angioedema is characterized by rapid development of **edematous swelling** involving skin and subcutaneous and submucosal tissues in various parts of the body.

Clinical Picture:

- **Rapidly developing edema** appear in less than half an hour and may last several days.

- **There is severe edema** of:-
  - The face and lips,
  - Circum-oral tissues,
  - Around the eyes,
  - Chin,
  - Tongue and also may affect the parotid.

- **This swelling appears:-**

  smooth, diffuse, **painless** with **no change in temperature** of the affected part.

  - Often there is pruritis.

- **Swelling of the nasopharynx and larynx** may produce respiratory emergency.

- **Types and etiology:**

  (1) Hereditary angioedema:

  This is a complement mediated angioedema which is a autosomal dominant trait.

  • **There are two types:-**

a- Type one HAE : low level of **Cl esterase inhibitor protein** of the plasma complement system **(quantitative deficiency) it represent *85%* of cases.**

b- Small number of cases are characterized by normal or even **increased level of non-functional Cl esterase inhibitor protein (qualitative deficiency) , it represent 15% of cases.**

This leads to continued complement activation with formation of kinin- like substance (probably C2b) that leads to increased capillary permeability and edema.

Affected individuals are highly susceptible **to laryngeal edema** leading to respiratory obstruction and death.

N.B.

In absence of Cl esterase inhibitor, under minimal stress, e.g. injury, emotional stress, trauma of dental procedures. uncontrolled activation of the classical pathway of complement may occur.

- **-C1 esterase inhibitor inhibits & regulate several plasma enzyme system including:-**

a. **The kinin** system.

b. **The fibrinolytic** enzyme system.

c. **The coagulation** system
(Hageman factor XII and factor XI).

-**Absence of C1 inhibitor** lead to formation of kinins with the resultant increase in vascular permeability and angioedema.

Precipitated by :-

- Blunt injury,

- Trauma of dental procedures

- Emotional stress.

- Also, may be precipitated by injection of drugs.

Management:

The concept is to supply the patient with **Cl esterase** inhibitor or **raise its level.**

1 - **Prophylactic treatment is now available:**

a- Plasma infusions**(fresh frozen plasma)** help in replacing the missing Cl esterase inhibitor activity.

b- **The synthetic androgen** "stanozolol": 2.5 - 10 mg daily, but should be avoided in liver disease since it causes cholestasis (stoppage or suppression of bile flow).

c- **The plasmin inhibitors:** aminocaporic acid and its analogue tranexamic acid (can inhibt kinin system) are useful preoperatively.

2 - Trauma and anxiety **should be avoided**.

Angeoedema

(2) IgE mediated angioedema	(3) Idiopathic angioedema
These are IgE dependent. They are associated with high levels of IgE and low IgA levels. They may represent atopic diathesis with specific antigen sensitivity (drugs, foods, pollens), Frequently, it may be associated with urticaria.	Affect 5-10% of population. It is not characterized with abnormal IgE levels. It may be precipitated by direct mast cell releasing agents. This can be induced by trauma, and drugs as opiates, antibiotics, curare and contrast media. Drugs may directly affect mast cell, causing the release of chemical mediators. The reaction may also be the result of over dose, toxicity or side effects of the drugs.

Mechanism

The development of swelling appears to be due to vasodilatation and increased capillary permeability brought about by histamine or histamine like substance with subsequent transudation of plasma in the subcutaneous and submucosal connective tissue of various parts of the body.

Treatment

a- Identification and removal of the cause. This may be difficult

b- Immediate relief of swelling particularly if larynx is involved by S.C injection of 0.5-1.0 ml of 1/1000 adrenline.

c- less server attacks can be managed by I.M injection of antihistamines (promethazine 25-50 mg or 10mg chiorpheniramine followed by adequate oral dose).

d- Corticosteroid in severe cases.

e- If this fails emergency tracheotomy may be life saving.

Hay fever

Definition:

It is a recurrent form of **allergy to wind born pollen grains, certain food or dust.**

It is seasonal, i.e. common in spring and early summer.

It appears to have **familial predisposition.**

Mechanism:

It is due to antigen antibody (IgE) reaction on the surface of mast cells with the release of histamine and other mediators in mucous membrane of upper respiratory tract and conjunctiva.

Clinical feature:

- Itching in the nose and profuse watery nasal discharge.
- Paroxysms of sneezing and coughing
- Conjuctival irritation and lacrimation
- Headache and depression.

- **Treatment:**

1- **Antihistaminic**

2- **Steroid** in refractory cases

3- **Desensitization**: antigen is administered on weekly basis at **low dose** several weeks before pollen season.

4- Avoid contact with the causative pollen **(travel to sea-shore).**

(4) Bronchial asthma

Definition:

It is an allergic disease characterized by recurrent **attacks of acute breathlessness.**

Mechanism:

-The patient is abnormally **sensitive to various substances inhaled or ingested.**

-Allergic reaction result in **release of leukotrienes (SRS-A)** which is the main mediator in bronchial asthma, resulting in: -

1 .**Spasm** of the smooth muscles of the bronchi.

2. **Swelling** of the bronchial mucosa.

3 .**Exudates** of mucous,

This will interfere with the flow of air during flow of air during inspiration and expiration

- **Clinical features:**

Sudden onset of:

1. Short forceful inspiration aided by accessory muscles of respiration.

2. Long easily heard wheezing expiration.

3. Expiration is not capable of deflating the chest.

4. Face is pale, nose, lip and ear are cyanotic.

5. The attack may last from **half an hour** to several hours.

Very prolonged attack is called Status asthmaticus

- **Dental implication**

1. **Assess the severity** of bronchial asthma.

2. What is current medical treatment, and the patient **should bring his medicine in every dental** appointment.

3. **Avoid dental treatment** if the medical status is not under control

4. **LA is better than GA**. Nitrous oxide and oxygen is completely safe.

5. **Avoid aspirin** (increase leukotreins release), **barbiturates and narcotics** (respiratory depressants)

6. **Oral candidosis** involving the soft palate and oropharynx is **common** due to the use of steroid inhaler.

7. **Steroid supplementation** may be needed if the patient is on systemic steroid or has had systemic steroid during the past year.

Treatment:

1- **Stop** dental procedure.

2- Check that **airway is clear**.

3- **Loosen** tight cloths.

4- Permit the patient to **sit upright**.

5- Ensure that the room is **airy and cool**.

6- The patient may **carry self medication** as;

- aerosol containing isoprenaline or adrenaline.

- **sublingual tablets** of isoprenaline.

7- 0.5 ml adrenaline s.c. or i.m.

8- Status asthmaticus

- Hospitalization,
- Oxygen inhalation.
- corticosteroids.
- **Sedation** the use of sedative drugs in severe or prolonged asthmatic attacks are absolutely contraindicated because they **depress the central nervous and respiratory systems.**

- **N.B.**

  1 - Antihistaminic are of no value, where Leukotrienes and not histamine are the important mediators of reaction.

  2 - **Antihistaminic are slowly acting** so they are of no value in emergency.

Urticaria

Definition:

It is a common allergic reaction on skin varying from **small localized papules** to extensive large coalescing plaque which is erythematous and characterized by itching.

Mechanism:

This type of allergy is due to:-

- Drugs,
- Food
- Certain bacteria that react with 1gE on mast cells resulting in the release of histamine and other mediators acting on superficial vasculature.

- **Treatment:**
  - Removal of the causative factor
  - Antihistaminics.
  - Corticosteroids.
  - Management of septic foci, since chronic Urticaria may be related to septic focus.

Type II hypersensitivity reaction

(Cytotoxic reaction)

This includes circulating cells, e.g. **R.B.C's, WBC's and platelets** whose surface serve as the antigen, that react with circulating antibodies in the presence of the complement causing cellular damage or lysis **(aplastic anemia, agranulocytosis, thrombocytopenia or hemolytic anemia).**

Thus, oral manifestations vary and may include oral infection, - hemorrhage, petechiae, pallor, atrophy of the lingual mucosa or oral ulceration, depending on the type of the affected blood forming element.

Type III hypersensitivity reaction

(1) Serum sickness

Definition:

It is a self limiting systemic allergic disease caused by immune-complex complement dependent reaction after injection of protein (foreign serum) or haptenic drug (penicillin).

Mechanism:

• Following a single large dose of injected antigen, there is slow degradation of antigen over several days , and the primary antibody response is initiated.

• 7 to 10 days after the injection, the produced antibodies will react with the residue of the circulating antigen (foreign serum) forming antigen-antibody complexes in small blood vessels in various organs.

• Antigen-antibody complexes result in complement fixation, attracting leukocytes, causing direct tissue damage (particularly vasculitis and arthritis).

- **Clinical manifestation:**

I. **Itching** followed by urticarial rash.

2. **Fever**.

3. **Generalized lymphadenopathy**.

4. **Swelling** of joints leading to arthralgia.

5. **Edema** in face, hands and feet.

6. **Asthma and rhinitis** may occur.

7. **Peripheral neuritis**, kidney disease and myocardial ischemia are less common features.

Treatment:

I. Corticosteroid is the drug of choice.

2. Antihistaminics for urticarial & edematous lesion but ineffective for arthralgia and fever.

N.B.

Serum accident is type I hypersensitivity reaction which is severe and fatal, developed in highly sensitized individuals on re-administration of foreign serum.

Its manifestation and management are similar to anaphylactic shock.

(2) Stomatitis medicamentosa (Allergic stomatitis)

Definition:

Allergic stomatitis are **eruptive lesions** of the oral mucosa resulting from **systemic administration** of allergen (drugs or food).

Mechanism:

- Allergic stomatitis are mediated by **humeral antibodies.**

It may be attributed to **type I (less common)** or **type III (more common)** hypersensitivity reaction.

- The allergen is administrated via injection, ingestion or percutaneous (absorption through skin).

Oral manifestation:

The allergic manifestation may occur in the **mouth alone or skin alone or may involve both.**

The oral manifestation are less common than that the cutaneous lesion (dermatitis medicamentosa).

1. **Acute multiple vesicles** and ulcers similar to erythema multiforme.

2. **Angioedema** of lip and oral mucosa.

3. **Urticaria** of lip.

4. Fixed drug eruption:-

- The eruptions appear at **the same site** whenever the drug is taken

- The drug is conjugated with the body protein in the vascular bed supplying a particular area.

- **The tongue is a common site** for fixed drug eruption.

Type IV hypersensitivity reaction

Stomatitis venenata / Dermatitis venenata (contact allergy)

Definition:

- **It is a delayed allergic reaction** of skin or oral mucosa to local contact with foreign substance (allergen).

- **The major immunologic features include:-**
  1. Mediated by TH1 and Tc cells.
  2. Caused by haptenic chemical.
  3. **Positive patch test.**

Basis of reaction:

Allergic contact stomatitis is mediated by **TH1 cells and Tc** may be involved.

In this process, the langerhans cells, mucosal macrophages, play a major role in recognition, processing and presentation of antigen to T cells Sensitization on **primary contact** takes several days, but once occur it lasts for years if not for life because of development of memory cells.

Second contact leads to release of sensitized cells which are able to proliferate and **release lymphokines.**

Lymphokines produce tissue damage via:-

- **Tumor necrosis factor β (lymphotoxin)** : non specifically kill some cells
- **Chemotactic factor**: attract macrophages which further secrete various kinds of monokines aiding in tissue damage.

The oral mucosa is less liable to sensitization compared to skin due to:

a) Brief contact

b) **Saliva**: dilutes, digests and washes away allergen

c) **High vascularity** of submucosa which leads to rapid adsorption and dispersion of allergens.

* While in skin: the keratin layer provides a good supply of protein to combine with hapten forming antigen complex.

Etiology: contact stomatitis may be induced by:

1) Induced by dentist: (iatrogenic)	2) Induced by patient:
A) Dental materials **1.Plastic filling material (acrylic resin) due to preparations: dentifrices, sensitization of the oral mucosa to liquid methyl denture powders & mouth methacrylates. washes.** **2- Denture base material due to liquid monomer.** **3- Amalgam filling** **B) Dental therapeutics:** **1-Antiseptic lozenges.** **2- phenol, alcohol..** **3- Topical antibiotics** **4- Iodine , formaline.**	**A) Dental cosmetic** **B) Chewing gum** **C) lip sticks.** **D) Food:** **1- candy** **2- melon** **3- shell-fish.**

Clinical features:

Stomatitis venenata:

The oral reaction may be **mild or severe** and usually localized in the form of: -

1. **Burning and itching** at the site of contact

2. **followed by erythema**, edema, vesicles or ulcers.

3. localized **atrophy of tongue coating**

4. Cheilitis (dryness, scaling of lip)

5. **Angular cheititis**

- **N.B.** edema and erythema involve denture bearing and denture contacting surfaces in contact allergy.

- It should be differentiated from denture sore mouth (chronic atrophic candidosis) in which edema and erythema are confined to denture bearing area only and patch test is negative.

- **Dermatitis venenata:**
  • Perioral skin may show itching, burning, erythema vesicles or urticarial rashes.

  • Dentist's or technician's fingers may also be affected when handling dental materials.

Patch test:

A) Dermal test	B) Epimucous test
The allergen (e.g. scrapping of denture) is applied to the skin with moist gause on one arm and control moist gauze alone on the other Allergic individuals show wheel reaction within 24-72 hours at the site of contact with	Application of allergen to oral mucosa via:- 1- Rubber suction cup applied to cheek mucosa. 2- Incorporation into orabase in other. mucobuccal fold. 3- Incorporation into oral adhesive spray. **4- Placing** it on palatal aspect of denture. allergen. *5-* Covering all palatal surface of denture with tin foil except small area to test sensitivity.

Allergic person **(positive test)** shows erythema, slough or ulceration at the site of contact within 24 - 72 hours.

Diagnosis of contact allergy:

I .History of contact with allergen.
2.Clinical examination: erythema, edema, vesicle & ulceration.
3.Positive patch test.

Treatment:

1.Mild: elimination of the causative agents.

2. Sever.

a. antihistaminics.
b. steroid (topical or systemic).

3. Symptomatic

a. antiseptic agent to minimize secondary infection.
b. topical anesthetic to relief pain.

D.D allergy to denture base material & denture sore mouth (DSM)

allergy to denture base material y.	Candidosis (DSM)
History: -Mostly new denture.	-Mostly old denture for a long time without relining or rebasing -Inadequate cleaning of the fittings surface of the denture. -Wearing of the denture for day and night.
Clinical examination: -Erythema is seen under the upper & lower denture (although more in upper). -Erythema involve the denture bearing & denture contacting areas (inner aspect of lip, check & tongue)	-Erythema is seen confined to the upper denture bearing area only. -Commonly seen confined to the upper denture not the lower.
Special investigations: -Patch test is +ve	1 .Patch test is —ye 2.Bacterial smear for direct examination of candida. 3 .Bacterial culture using medium. 4.High candidal antibody titre in serum & saliva.
Therapeutic test -Antihistaminic	-Antifungal drugs.

A comparison between allergic stomatitis and contact stomatitis

	Allergic stomatitis	Contact stomatitis
1.Route of administration	Systemic (injection ,ingestion or absorption across skin and/or oral mucosa)	Topical (adsorption)
2. Type of hypersensitivity	Humoral a) Type III (common) b) Type I (less common)	Cellular Type IV
3. Site of sensitization	In the CT, sometimes in certain vasuclar network (fixed drug eruption)	Result of sensitization of epidermis.
4. Site of lesions produced	Anywhere on the body	Only at site of contact
5. Skin test for allergen	Patch test, weak positive	Positive patch test
6. Onset	Early onset within few hours	Appear later (48 hrs)
7. Severity	More severe	Less severe